

Generation Interconnection System Impact Study Report

For Queue Project Q22-2101

Churchtown 138 kV

April 15, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2101. The Interconnection Customer proposes a 260.8 MW solar photovoltaic facility interconnecting at the Churchtown 138 kV substation in the New Jersey area. The request was received on May 31, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 737 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$45,045,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Gentilly Road - Churchs Ferry 138 kV
Contingency: Chambers 345-138 kV Transformer Bank
Loading: 206.2% (Rating: 201 MVA)
DFAX: 73.6%
Pre-project loading: 110.8%
2. Facility: Denham Springs Substation - Alexander St 138 kV
Contingency: Chambers 345 kV/138 kV Auto
Loading: 155.9% (Rating: 199 MVA)
DFAX: 9.5%
Pre-project loading: 143.5%
3. Facility: Keney - Dell Crk 138 kV
Contingency: Red Lion Energy Ctr - Delaware City Plant Substation 138 kV Line
Loading: 144.8% (Rating: 288 MVA)
DFAX: 13.4%
Pre-project loading: 132.7%
4. Facility: Mill Cteek - Southpoint 138 kV
Contingency: Chambers 345-138 kV Auto
Loading: 142.6% (Rating: 172 MVA)
DFAX: 7.4%
Pre-project loading: 131.4%
5. Facility: Sunset Lake - Keeney 138 kV
Contingency: Red Lion Energy Ctr - Delaware City Plant Substation 138 kV
Loading: 133.7% (Rating: 242 MVA)
DFAX: 13.4%
Pre-project loading: 119.3%
6. Facility: Churchtown - Chambers 138 kV
Contingency: Churchs Ferry - Gentilly Rd 138 kV
Loading: 128.7% (Rating: 253 MVA)

- DFAX: 81.3%
Pre-project loading: 45.1%
7. Facility: Sunset Lake - Broadview 138 kV
Contingency: Chambers 345 kV/138 kV Transformer Bank
Loading: 125.2% (Rating: 195 MVA)
DFAX: 20.2%
Pre-project loading: 98.1%
8. Facility: Churchtown - Gentilly Rd Substation 138 kV
Contingency: Chambers 345/138 kV Transformer Bank
Loading: 112.4% (Rating: 224 MVA)
DFAX: 73.6%
Pre-project loading: 26.7%
9. Facility: Dell Creek - Churchs Ferry Substation 138 kV Line
Contingency: Chambers 345-138 kV Autotransformer
Loading: 107.6% (Rating: 286 MVA)
DFAX: 48.8%
Pre-project loading: 63.1%
10. Facility: Maurice - Red Lion Energy Ctr 138 kV
Contingency: Keeney - Dell Creek 138 kV line
Loading: 107.1% (Rating: 230 MVA)
DFAX: 8.6%
Pre-project loading: 97.4%
11. Facility: Delaware City Plant - Red Lion Energy Center 138 kV Line
Contingency: Keeney - Dell Creek 138 kV
Loading: 102.4% (Rating: 273 MVA)
DFAX: 8.6%
Pre-project loading: 94.2%
12. Facility: Harmony - Broadview 138 kV
Contingency: Tombstone - Bixby Hill 500 kV
Loading: 101.8% (Rating: 256 MVA)
DFAX: 6.5%
Pre-project loading: 95.2%
13. Facility: Southpoint Substation - Elizabeth Lake 138 kV
Contingency: Chambers 345/138kV TX
Loading: 101.7% (Rating: 206 MVA)
DFAX: 7.4%
Pre-project loading: 92.3%
14. Facility: Chambers 345-138 kV TX
Contingency: Gentilly Road - Churchs Ferry 138 kV
Loading: 100.6% (Rating: 456 MVA)
DFAX: 81.3%
Pre-project loading: 54.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gentilly Rd - Churchs Ferry 138 kV	Rebuild line with higher-capacity conductor	\$10,859,000	66% (\$7,195,000)
Denham Springs Substation - Alexander St 138 kV	Rebuild line with higher-capacity conductor	\$10,086,000	66% (\$6,681,000)
Keney - Dell Crk 138 kV	Rebuild line with higher-capacity conductor	\$2,837,000	39% (\$1,118,000)
Mill Cteek - Southpoint 138 kV	Rebuild line with higher-capacity conductor	\$3,724,000	66% (\$2,467,000)
Keeney - Sunset Lake 138 kV	Rebuild line with higher-capacity conductor	\$2,034,000	39% (\$802,000)
Churchtown - Chambers 138 kV line	Rebuild line with higher-capacity conductor	\$3,004,000	93% (\$2,797,000)
Sunset Lake - Broadview 138 kV	Rebuild line with higher-capacity conductor	\$3,827,000	66% (\$2,536,000)
Churchtown - Gentilly Rd Substation 138 kV	Rebuild line with higher-capacity conductor	\$7,569,000	66% (\$5,015,000)
Dell Creek - Churchs Ferry Substation 138 kV Line	Reconductor line	\$1,143,000	58% (\$660,000)
Maurice - Red Lion Energy Ctr 138 kV	Reconductor line	\$967,000	41% (\$395,000)
Delaware City Plant - Red Lion Energy Center 138 kV Line	Reconductor line	\$982,000	41% (\$401,000)
Harmony - Broadview 138 kV	Reconductor line	\$793,000	51% (\$402,000)
Elizabeth Lake - Southpoint 138 kV	Reconductor line	\$586,000	66% (\$388,000)
Chambers 345-138 kV TX	Replace transformer with higher-rated unit	\$22,026,000	64% (\$14,188,000)
	Total allocated to project		\$45,045,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.